

Transfer Function Representation in MATLAB

Two ways to build the same transfer function object:

1. `zpk()` from the ratio of factors
2. `tf()` from the ratio of polynomials.

Converting one form to the other checks that both describe the same system.

$$G(s) = \frac{3}{s^3 + 9s^2 + 20s} = \frac{3}{s(s+4)(s+5)}$$

Ratio of factors: zeros, poles, gain.

```
Gz = zpk([], [0 -4 -5], 3)
```

Ratio of polynomials: coefficient vectors in descending powers of s .

```
Gt = tf(3, [1 9 20 0])
```

Check by converting one form to the other.

```
tf(Gz)
```

$$H(s) = \frac{10(s+2)}{s^2(s+4)(s+5)}$$

```
Hz = zpk(-2, [0 0 -4 -5], 10)
```

The denominator is a product of factors. `conv()` multiplies two polynomials.

```
den = conv([1 0 0], conv([1 4], [1 5]))  
Ht = tf(10*[1 2], den)
```

Check.

```
zpk(Ht)
```